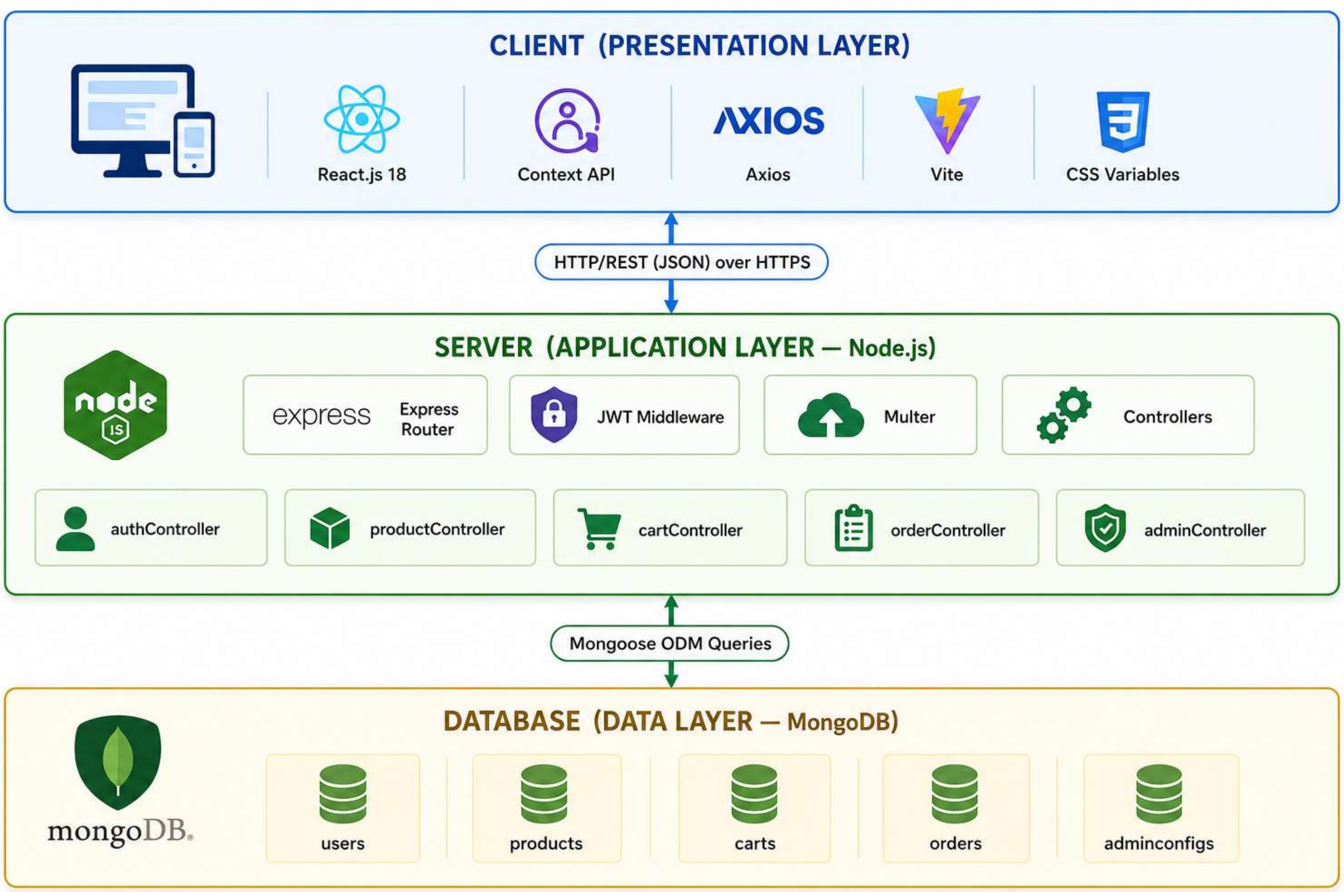
# DOCUMENT: TECHNOLOGY STACK

| Phase | Project Design Phase II |
|---|---|
| Document | Technology Stack (Architecture & Stack) |
| Project Name | **PetPaws – Pet Foods & Accessories Store** |
| Team ID | [TEAM-ID] |
| Date | 14 June 2026 |
| Maximum Marks | 4 Marks |

# Technology Stack (Architecture & Stack)

## Technical Architecture Overview

PetPaws follows a 3-tier MERN stack architecture: Presentation Layer (React.js SPA), Application Layer (Node.js + Express.js REST API), and Data Layer (MongoDB + Mongoose ODM).



## Table 1: Components & Technologies

| S.No | Component | Description | Technology |
|---|---|---|---|
| 1 | **User Interface** | Single Page Application (SPA) served via browser on desktop and mobile | React.js 18 (Vite 5 build tool) |
| 2 | **Client-Side Routing** | Declarative navigation with protected routes for authenticated and admin-only pages | React Router v6 |
| 3 | **Global State Management** | AuthContext for JWT user state; CartContext for real-time cart count and totals | React Context API + localStorage |
| 4 | **HTTP Client** | Centralised Axios instance with JWT request interceptor and 401 response interceptor | Axios 1.x |
| 5 | **Styling** | CSS Custom Properties (design tokens), responsive grid, mobile-first breakpoints | Pure CSS3 with CSS Variables |
| 6 | **Notifications** | Non-blocking toast alerts for cart actions, authentication, and order events | react-hot-toast 2.x |
| 7 | **Icons** | Lightweight SVG icon set used across Navbar, cart, profile, and admin UI | react-icons (Feather set) |
| 8 | **API Server / Application Logic** | RESTful API handling all business logic: auth, products, cart, orders, reviews, wishlist, admin | Node.js (v18+) + Express.js 4 |
| 9 | **Authentication Logic** | JWT generation, verification middleware (protect), role authorization middleware (adminOnly) | jsonwebtoken + bcryptjs |
| 10 | **Database** | Document-oriented NoSQL database for Users, Products, Orders, Cart, Reviews, AdminSettings collections | MongoDB 7.x (local) / MongoDB Atlas (cloud) |
| 11 | **Database ODM** | Schema definition, validation, pre-save hooks (password hashing, discount computation), text indexes, aggregation pipeline | Mongoose ODM v8 |
| 12 | **Environment Configuration** | Stores PORT, MONGO_URI, JWT_SECRET, JWT_EXPIRE; promotes from dev to prod without code changes | dotenv 16.x |

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| 13 | **Development Server & Proxy** | Fast HMR in development; proxies /api/* to localhost:8000 to eliminate CORS issues in dev | Vite 5 dev server |
| 14 | **Version Control** | Source code management, branching strategy, and collaborative development | Git + GitHub |
| 15 | **API Testing** | Manual API endpoint testing and documentation during development | Postman |
| 16 | **Code Editor** | Primary development IDE with integrated Git, terminal, and extension ecosystem | Visual Studio Code |

## Table 2: Application Characteristics

| S.No | Characteristic | Description | Technology |
|---|---|---|---|
| 1 | **Open-Source Frameworks** | All technologies used in PetPaws are open-source with active community support: React (Meta), Express.js (Node.js Foundation), MongoDB (SSPL / Apache 2.0), Mongoose (MIT), Vite (MIT), Axios (MIT) | React 18, Express 4, MongoDB, Mongoose 8, Vite 5, Axios |
| 2 | **Security Implementations** | Password storage: bcrypt SHA-256 hashing with 10 salt rounds. Token security: JWT HS256 algorithm with 7-day expiry. API protection: JWT middleware on all private routes. Role enforcement: adminOnly middleware for admin routes. Input validation: Mongoose schema validators. Soft delete: isActive flag prevents data loss. | bcryptjs, jsonwebtoken, Mongoose validators, CORS |
| 3 | **Scalable Architecture** | Three-tier architecture (Presentation → API → Database) enables independent horizontal scaling of each layer. RESTful stateless API supports load balancing without shared session state. MongoDB Atlas auto-sharding supports petabyte-scale growth. Modular Express route/controller structure enables microservice decomposition. | 3-tier MERN, MongoDB Atlas sharding, stateless JWT |
| 4 | **Availability** | Decoupled deployment: frontend (Vercel CDN), backend (Railway/Render auto-restart), database (MongoDB Atlas 99.995% uptime with replica set failover). Stateless JWT authentication | Vercel, Railway/Render, MongoDB Atlas Replica Set |

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| | | eliminates single-point-of-failure session stores. Vite proxy in development; environment-driven base URL in production for zero-downtime migrations. | |
| 5 | **Performance** | API pagination: 12 products/page, 20 orders/page reduces response size. MongoDB text index on (name, description, brand, tags) enables fast full-text search. Server-side aggregation pipeline for dashboard KPIs avoids large client-side data processing. Vite production build with code splitting and tree-shaking minimises JavaScript bundle size. React state management avoids unnecessary re-renders via Context isolation. | MongoDB text index, aggregation pipeline, Vite code splitting |